Joseph Tran

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Education

University of Nevada, Reno School of Medicine — Reno, NV
M.D./Ph.D. Candidate in Integrative Neuroscience — *Aug. 2023 – 2030, anticipated*

Stanford University — Stanford, CA
Bachelor of Science in Computer Science — *Aug. 2018 – March 2023*

Honors and Awards

2026: AWE Reality Hack Finalist, MIT Reality Hack x Snap
- Top 4 of 20+ competing teams

2025: Jerry and Janet Zebrack Family Endowed Scholarship for Humanism, University of Nevada, Reno School of Medicine (UNR SOM)
- Nominated by classmates and chosen by an official selection committee for demonstrating humility, respect, and humanistic values in patient care and teamwork
- Recognized with a $4,500 prize

2024: Student Outreach Clinic Summer Project Grant, UNR SOM
- Awarded $500 to pursue a research project through the Student Outreach Clinic

2023: 1st Place Poster Presentation, Inaugural Wolf Pack Pediatric Symposium
2023: AOA Grant, AOA
- Awarded $1,000 to create winter kits for homeless individuals throughout Reno.

Research

Human Machine Perception Lab, UNR — Reno, NV
Graduate Research Assistant; Advisor: Alireza Tavakkoli, Ph.D. — *Oct. 2023 – Current*
- Developing suite of visual ophthalmic tests in virtual reality to better understand visual perception.
- Implementing Vision Suite of tests in Student Outreach Clinic for feasibility study. Included IRB submission.
- Collaborating with NASA to develop VR platform for monitoring eye health in microgravity to better understand Spaceflight-Associated Neuro-Ocular Syndrome (SANS)

Stanford Byers Eye Institute — Stanford, CA
Research Assistant; Advisor: Sophia Wang, M.D. — *Oct. 2023 – Current*
- Conducted research on glaucoma detection in OCT images using RETFound, a deep learning model.
- Preprocessed and analyzed medical imaging data to train and evaluate model performance.
- Assisted in developing and optimizing machine learning pipelines for automated glaucoma diagnosis.

Visualization Lab, Wu Tsai Institute — Stanford, CA
Stanford Computer Science Honors Thesis Research — *Aug. 2021 – Mar. 2023*
- Designed virtual reality Disparometer application to test whether VR headsets are a viable method for conducting vision research on patients affected with strabismus.
- VR Application will be tested on clinical patients and results will be presented to panel of Stanford Professors.

BAYER Pharmaceuticals — St. Louis, MO (remote)
Software Engineer — *Jun. 2021 – Sep. 2021*
- Developed virtual reality training program to help Bayer operators learn complex equipment terminology.
- Reduced polycount by 90% of 3D modeled equipment.

Incubator for Medical Mixed Reality (IMMERS) Lab — Stanford, CA
Undergraduate Research Assistant — *June 2019 – June 2020*
- Expanded upon traditional methods of medicine and healthcare by leveraging mixed reality technology.
- Developed an augmented reality (AR) application using the Microsoft Hololens 1, visualizing scattered x-ray to improve education and awareness of scattered radiation in interventional suites. Resulted in a poster presentation.
- Creating an AR application to aid Transcranial Magnetic Stimulation (TMS) technicians by visualizing EEG fields of stimulated brain areas using precomputed MRI patient data and Matlab algorithms.

Publications

1. **Tran, J.,** Ravindranath, R., Chang, R. T. & Wang, S. Y. Evaluating RetFound: A Foundation Model for Glaucoma Detection Using Macular OCT Images. *Investigative Ophthalmology & Visual Science* **66,** 391 (June 30, 2025).
2. Maturi, J., **Tran, J.,** Boucher, N., Winsor, J. & Scott, A. Utilization Patterns and Adverse Outcomes of Anti-Complement Therapy in Geographic Atrophy in the Clinical Setting. *Investigative Ophthalmology & Visual Science* **66,** 666 (June 30, 2025).
3. **Tran, J.,** Pasia, G., Flores Sanchez, D. & Westhoff, J. Abstract HUP2: Geographical Trends in Cerebrovascular Disease Mortality in the United States, 1999-2020. *Stroke* **56,** AHUP2–AHUP2 (Jan. 30, 2025).

Currently Under Review

Tran J et al. (2026) "Evaluating RETFound: A Foundation Model for Glaucoma Detection Evaluated using Macular Optical Coherence Tomography Images"

Maturi J, Tran J et al. (2026) "Early Utilization Patterns and Adverse Outcomes of Anti-Complement Therapy in Geographic Atrophy in the Clinical Setting"

Presentations

Tran J. *Designing a Virtual-Reality Neuro-Visual Assessment Platform for Spaceflight-Associated Neuro-Ocular Syndrome.* Speaker Presentation at Microgravity to Medicine Symposium, International VR Healthcare Association (IVRHA); Orlando, Florida; October 27, 2026.

Tran J, Ravindranath R, Chang R, Wang S. *Evaluating RetFound: A Foundation Model for Glaucoma Detection Using Macular OCT Images.* Poster Presentation for ARVO Annual Meeting; Salt Lake City, Utah; May 5, 2025.

Maturi J, **Tran J**, Boucher N, Winsor J, Scott A. *Utilization Patterns and Adverse Outcomes of Anti-Complement Therapy in Geographic Atrophy in the Clinical Setting.* Poster Presentation at for ARVO Annual Meeting. Poster awarded MIT Outstanding Poster for section; Salt Lake City, Utah; May 5, 2025.

Tran J, Pasia LG, Flores D. *Geographical Trends in Cerebrovascular Disease Mortality in the United States, 1999-2020.* Selected top 20 abstract for HEADS UP Symposium, International Stroke Conference; Los Angeles, CA; February 4, 2025.

Tran J. *RETFound Evaluation for Glaucoma Detection Using OCT Images.* Selected oral presentation for Medical Student Research Day; University of Nevada, Reno School of Medicine; November 20, 2024.

Pasia LG, **Tran J**, Flores D. *Trends and Disparities in Firearm-Related Mortality Among US Children and Young Adults, 1999-2020.* Top poster presentation at Inaugural Wolf Pack Pediatric Symposium; University of Nevada, Reno; May 23, 2024.

Service, Leadership, Experience

University of Nevada, Reno School of Medicine Reno, NV

Ophthalmology Student Interest Group Leader *Dec. 2024 – May 2025*

- Facilitate networking opportunities to learn about ophthalmology and connect with local ophthalmologists.

Street Medicine Student Outreach Clinic Manager *Dec. 2024 – May 2025*

- Led the integration of the Street Medicine Student Interest Group into the established Student Outreach Clinic.
- Developed and implemented a structured schedule to ensure regular clinic operations.
- Successfully organized and operated 9+ clinics, reaching hundreds of homeless individuals. Facilitated medical care for dozens of patients and referred numerous individuals to the free Student Outreach Clinic.

Street Medicine Student Interest Group Leader *Dec. 2024 – May 2025*

- Secured funding and donations to acquire essential supplies for individuals experiencing homelessness.
- Organized events where students assembled care kits for distribution to the homeless community.
- Facilitated partnerships with local organizations to maximize outreach and impact.

Eddy House Reno, NV

Life Skills Instructor *Dec. 2023 – May 2025*

- Teach classes on first aid, health education, and essential life skills to 50+ youth at risk for homelessness.
- Developed and implemented a structured schedule to ensure regular clinic operations.
- Develop and deliver engaging lessons on topics such as personal safety, hygiene, and emergency preparedness.

Stanford XR (Augmented and Virtual Reality) Club Stanford, CA

Vice President, Internal *Sep. 2018 – Jun. 2022*

- Lead community outreach efforts including planning social events and running weekly meetings and managing newsletter to bring awareness and excitement of XR to local community.

Leader of Project Cohort and Community Outreach *Sep. 2019 – Jun. 2020*

- Manage and lead 20+ students interested in Virtual and Augmented Reality (XR). Help students form project teams and pair them with experienced faculty/industry members to guide them through a year-long XR project.
- Took on the challenge of making XR more accessible to new community members, resulting in tens of new people joining and participating in the club.

Student Teacher for Augmented Reality Stanford Course: CS 12SI *Sep. 2020 – Dec. 2020*

- Teach 30+ students how to create mobile augmented reality apps using Unity AR Foundations.

Projects

Curiosity Kids [HACKATHON FINALIST] | *VR Development, Lens Studio, Snap* Jun. 2026

- Created during AWE Reality Snap Hack 2026. Finalist among 20+ teams.
- Using Snap Spectacles and Lens Studio, created a child-centered curiosity engine that encourages children to interact and learn from the world around them.
- Helped with background research, wrote white paper, developed business plan, write documentation and presentation.
- Created front-end UI elements and backend identification workflow.
- Hackathon Devpost

Strabismus Project | *VR Development, C#* Mar. 2023

- Developed a virtual reality application to measure subjective vs. objective measures in strabismus patients.

Operating Suite X-ray Visualizer | *Augmented Reality, Hololens, C#* Jun. 2020

- Developed an augmented reality (AR) application using the Microsoft Hololens 1, visualizing scattered x-ray to improve education and awareness of scattered radiation in interventional suites. Resulted in a poster presentation.

Stroke Rehabilitation with Mobile AR | *Augmented Reality, Unity, C#, Microsoft Azure* Feb. 2019

- Developed a mobile augmented reality application designed to encourage stroke rehabilitation by guiding users through hand exercises.

Professional Membership

2025 – Present Member, BRAIN Core, University of Nevada, Reno.

2024 – Present Member, Association for Research in Vision and Ophthalmology (ARVO).

2023 – Present Member, American Heart Association (AHA).

Technical Skills

Languages: Python, C/C++, C#
Developer Tools: Git, Google Cloud Platform, VS Code, Visual Studio, Jupyter Notebook, Unity, Lens Studio

Personal Interests

Rock Climbing, Social Dance, Piano